

UA1 Projet1 G4 Les_Hadoop_Riders

Projet 1 — Déploiement et Exploitation d'un Cluster Hadoop avec Docker

Cours	Bases de données massives avancées (IFM30522)
Unité	UA1 — Projet 1
Groupe	G4 — Les_Hadoop_Riders
Image Docker	leshadoopriders/hadoop-tp-g4:1.0
Docker Hub	https://hub.docker.com/r/leshadoopriders/hadoop-tp-g4
Code source	https://github.com/asinyopetro/Les_Hadoop_Riders_G4

Membres du groupe

Étudiant	Étudiant
Komla Petro Asinyo	Kassoum Dene
Joel Kazoni Tugirimana	Forbes Magène
Frank A Simo Ngounou	Wren Surprenant-Nicolson

Objectif du livrable : déployer un cluster Hadoop multi-nœuds avec Docker, manipuler HDFS, exécuter un job YARN, publier l'image sur Docker Hub, et documenter les résultats avec captures.

Partie 1 — Contexte théorique et architecture (5 pts)

1. Cas d'usage Big Data (1.5 pt)

Nous avons choisi le secteur de l'**e-commerce** (commerce électronique).

Exemple concret : une chaîne de magasins qui conserve les tickets de caisse, les logs du site web et les mouvements de stock. Chaque jour le volume grossit, les données arrivent souvent, et les formats changent (CSV, JSON, textes, images produits).

Les **3V** du Big Data dans ce contexte :

V	Signification	Application e-commerce
Volume	Quantité de données	Millions d'événements, historique sur plusieurs années
Vélocité	Vitesse d'arrivée	Commandes et paiements en continu ; réaction rapide (fraude, rupture)
Variété	Diversité des formats	Tables, logs, JSON, images produits

HDFS permet de stocker ces données de façon distribuée. **YARN** permet de lancer des traitements (MapReduce, etc.) sur le cluster.

2. HDFS vs système de fichiers classique (1.5 pt)

Critère	FS classique (NTFS, ext4...)	HDFS
Emplacement	Une seule machine	Plusieurs machines (cluster)
Organisation	Fichier entier sur un disque	Fichier découpé en blocs
Métadonnées	Gérées par l'OS local	Gérées par le NameNode
Tolérance aux pannes	Faible (disque unique)	Forte grâce à la réplication
Contenu des fichiers	Sur le même disque que les métadonnées	Sur les DataNodes

Sur un système local, un fichier vit sur **une** machine. Avec **HDFS**, le fichier est coupé en blocs, répartis et répliqués sur plusieurs DataNodes. Le NameNode conserve le namespace (noms, dossiers, emplacement des blocs).

En pratique : le FS local est centralisé ; HDFS est distribué et plus tolérant aux pannes grâce à la réplication.

3. Rôle de YARN — ResourceManager et NodeManager (2 pts)

YARN (*Yet Another Resource Negotiator*) gère les ressources CPU/mémoire du cluster lorsqu'une application est soumise.

Composant	Où ?	Rôle
ResourceManager (RM)	Master	Reçoit la demande, connaît les ressources, alloue des conteneurs, démarre l'ApplicationMaster
NodeManager (NM)	Chaque worker	Démarre et surveille les conteneurs sur sa machine , remonte l'état au RM
ApplicationMaster	Conteneur alloué	« Chef » du job : suit l'exécution Map/Reduce

À retenir : le ResourceManager **orchestre** ; le NodeManager **exécute** localement.

Partie 2 — Déploiement et manipulation pratique (15 pts)

A. Mise en place de l'environnement Docker (4 pts)

1. Déploiement NameNode + 5 DataNodes (1.5 pt)

Nous avons déployé **six conteneurs** sur le réseau Docker `hadoop-net` :

Conteneur	Rôles
hadoop-master	NameNode, ResourceManager, SecondaryNameNode
hadoop-worker1 ... hadoop-worker5	DataNode + NodeManager

Ports exposés :

Port	Service
9870	Interface web HDFS (NameNode UI)
8088	Interface web YARN
9000	HDFS RPC

```
docker compose up --build -d
docker ps
```

```
Windows PowerShell — Les_Hadoop_Riders_G4

PS C:\Users\asiny\Desktop\Les_Hadoop_Riders_G4> docker ps
NAME                STATUS              PORTS
hadoop-worker2      Up 10 hours         8042/tcp, 8088/tcp, 9000/tcp, 9864/tcp, 9870/tcp
hadoop-worker3      Up 10 hours         8042/tcp, 8088/tcp, 9000/tcp, 9864/tcp, 9870/tcp
hadoop-worker5      Up 10 hours         8042/tcp, 8088/tcp, 9000/tcp, 9864/tcp, 9870/tcp
hadoop-worker1      Up 10 hours         8042/tcp, 8088/tcp, 9000/tcp, 9864/tcp, 9870/tcp
hadoop-worker4      Up 10 hours         8042/tcp, 8088/tcp, 9000/tcp, 9864/tcp, 9870/tcp
hadoop-master       Up 10 hours         0.0.0.0:8088->8088/tcp, [::]:8088->8088/tcp, 0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp, 0.0.0.0:9870->9870/tcp, [::]:9870->9870/tcp

PS C:\Users\asiny\Desktop\Les_Hadoop_Riders_G4> docker exec -it hadoop-master hdfs dfsadmin -report
Configured Capacity: 5405505884160 (4.92 TB)
Present Capacity: 5032864239616 (4.58 TB)
DFS Remaining: 5032862023680 (4.58 TB)
DFS Used: 2215936 (2.11 MB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
-----
Live datanodes (5):

Name: 172.23.0.3:9866 (hadoop-worker5.les_hadoop_riders_g4_hadoop-net)
Hostname: hadoop-worker5
Decommission Status : Normal
Configured Capacity: 1081101176832 (1006.85 GB)
```

Capture terminal — docker ps / dfsadmin

2. Vérification des DataNodes (1.5 pt)

```
docker exec -it hadoop-master hdfs dfsadmin -report
```

Résultat observé : Live datanodes (5) — le cluster est opérationnel.
Interface NameNode : http://localhost:9870

Hadoop

Overview

Datanodes

Datanode Volume Failures

Snapshot

Startup Progress

Utilities

Overview

'hadoop-master:9000' (✓active)

Started:

Sat Oct 03 11:02:46 -0400 2026

Version:

3.3.6, r1be78238728da9266a4f88195058f08d012bf9c

Compiled:

Sun Jun 18 04:22:00 -0400 2023 by ubuntu from (HEAD detached at release-3.3.6-RC1)

Cluster ID:

CID-c8f2f423-c208-4cdf-9c10-5573319713da

Block Pool ID:

BP-329230766-172.23.0.2-1791033373712

Summary

Security is off.

Safemode is off.

32 files and directories, 9 blocks (9 replicated blocks, 0 erasure coded block groups) = 41 total filesystem object(s).

Heap Memory used 89.9 MB of 242 MB Heap Memory. Max Heap Memory is 2.4 GB.

Non Heap Memory used 69.58 MB of 72.25 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

Configured Capacity:

4.92 TB

Configured Remote Capacity:

0 B

DFS Used:

2.18 MB (0%)

Non DFS Used:

90.97 GB

DFS Remaining:

4.58 TB (93.11%)

Block Pool Used:

2.18 MB (0%)

DataNodes usages% (Min/Median/Max/stdDev):

0.00% / 0.00% / 0.00% / 0.00%

Live Nodes

5 (Decommissioned: 0, In Maintenance: 0)

Dead Nodes

0 (Decommissioned: 0, In Maintenance: 0)

Decommissioning Nodes

0

UI NameNode

Hadoop

Overview

Datanodes

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Datanode Information

✓ In service

⬇ Down

🔄 Decommissioning

🚫 Decommissioned

🛑 Decommissioned & dead

🟡 Entering Maintenance

🔴 In Maintenance

🛑 In Maintenance & dead

Datanode usage histogram

Disk usage of each DataNode (%)

In operation

DataNode State

All

Show

25

entries

Search:

Node

Http Address

Last contact

Last Block Report

Used

Non DFS Used

Capacity

Blocks

Block pool used

Version

✓/default-rack/hadoop-worker1:9866 (172.23.0.5:9866)

http://hadoop-worker1:9864

2s

258m

684 KB

18.19 GB

1006.85 GB

7

684 KB (0%)

3.3.6

✓/default-rack/hadoop-worker3:9866 (172.23.0.4:9866)

http://hadoop-worker3:9864

2s

354m

344.99 KB

18.19 GB

1006.85 GB

3

344.99 KB (0%)

3.3.6

✓/default-rack/hadoop-worker4:9866 (172.23.0.7:9866)

http://hadoop-worker4:9864

2s

263m

353.43 KB

18.19 GB

1006.85 GB

5

353.43 KB (0%)

3.3.6

✓/default-rack/hadoop-worker5:9866 (172.23.0.3:9866)

http://hadoop-worker5:9864

2s

278m

178.31 KB

18.19 GB

1006.85 GB

5

178.31 KB (0%)

3.3.6

✓/default-rack/hadoop-worker2:9866 (172.23.0.6:9866)

http://hadoop-worker2:9864

2s

278m

667.16 KB

18.19 GB

1006.85 GB

6

667.16 KB (0%)

3.3.6

3. Publication de l'image (1 pt)

```
docker login
docker push leshadoopridders/hadoop-tp-g4:1.0
```

URL publique : <https://hub.docker.com/r/leshadoopridders/hadoop-tp-g4>

Note : Docker Hub impose les minuscules, d'où `g4` plutôt que `G4` .

B. Manipulation avancée sur HDFS (6 pts)

4. Répertoires et droits (2 pts)

Fichier local préparé : `data/transactions.csv` (colonnes ID, Date, Montant, Magasin — 10 lignes).

```
hdfs dfs -mkdir -p /data/ventes/2026
hdfs dfs -chmod 755 /data/ventes/2026
hdfs dfs -put -f /data/transactions.csv /data/ventes/2026/
hdfs dfs -chmod 644 /data/ventes/2026/transactions.csv
```

5. Transfert et vérification des blocs (2 pts)

```
hdfs fsck /data/ventes/2026/transactions.csv -files -blocks -locations
hdfs dfs -stat "%n | taille=%b | replication=%r | block_size=%o"
/data/ventes/2026/transactions.csv
```

Indicateur	Valeur observée
Taille	377 octets
Réplication initiale	3
Nombre de blocs	1 (fichier petit ; bloc par défaut 128 Mo)
État FSCK	HEALTHY

```
root@hadoop-master ~# HDFS

root@hadoop-master:~# hdfs dfs -ls /data/ventes/2026
Found 2 items
-rw-r--r-- 2 hadoop supergroup 377 2026-10-03 13:30 /data/ventes/2026/transactions.csv
-rw-r--r-- 3 hadoop supergroup 567 2026-10-03 13:31 /data/ventes/2026/transactions_merged.csv
root@hadoop-master:~# hdfs dfs -cat /data/ventes/2026/transactions.csv | head -n 5
ID,Date,Montant,Magasin
T001,2026-01-05,49.99,Montreal-Centre
T002,2026-01-06,129.50,Laval
T003,2026-01-07,15.25,Quebec
T004,2026-01-08,250.00,Montreal-Ouest
root@hadoop-master:~# hdfs fsck /data/ventes/2026/transactions.csv -files -blocks -locations 2>&1 | head -n 25
Connecting to namenode via http://hadoop-master:9870/fsck?ugi=root&files=1&blocks=1&locations=1&path=%2Fdata%2Fventes%2F2026%2Ftransactions.csv
FSCK started by root (auth:SIMPLE) from /172.23.0.2 for path /data/ventes/2026/transactions.csv at Sun Oct 04 01:02:57 GMT 2026

/data/ventes/2026/transactions.csv 377 bytes, replicated: replication=2, 1 block(s): OK
0, BP-329230766-172.23.0.2-1791033373712:blk_1073741825_1001 len=377 Live_repl=2 [DatanodeInfoWithStorage[172.23.0.4:9866,D5-6b0bc5c8-0343-4441-977a-45186fdeff03,DISK], DatanodeInfoWithStorage[172.23.0.6:9866,D5-f042a204-27cf-4970-b237-39c216ef47fb,DISK]]

Status: HEALTHY
Number of data-nodes: 5
Number of racks: 1
Total dirs: 0
Total symlinks: 0

Replicated Blocks:
Total size: 377 B
Total files: 1
Total blocks (validated): 1 (avg. block size 377 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 0 (0.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 3
Average block replication: 2.0
Missing blocks: 0
Corrupt blocks: 0
```

Capture terminal — HDFS

6. Lecture et concaténation (1 pt)

```
hdfs dfs -cat /data/ventes/2026/transactions.csv | head -n 5
hdfs dfs -put -f /data/transactions_part2.csv /data/ventes/2026/
hdfs dfs -getmerge /data/ventes/2026 /tmp/transactions_merged.csv
hdfs dfs -put -f /tmp/transactions_merged.csv
/data/ventes/2026/transactions_merged.csv
```

Les 5 premières lignes sont lues directement depuis HDFS. Les deux fichiers ont été fusionnés via `getmerge`.

7. Suppression / corbeille (1 pt)

```
hdfs dfs -rm /data/ventes/2026/transactions_part2.csv
hdfs dfs -D fs.trash.interval=10080 -rm /data/ventes/2026/to_delete.csv
hdfs dfs -ls -R /user/hadoop/.Trash
# suppression définitive :
# hdfs dfs -rm -skipTrash /chemin/fichier
```

Sans `fs.trash.interval`, la suppression peut être définitive. Avec un intervalle > 0 , le fichier passe dans `.Trash`. L'option `-skipTrash` force la suppression définitive.

C. Exécution d'un job YARN (5 pts)

Exemple choisi : **calcul de π** par méthode de Monte Carlo (exemple officiel MapReduce).

```
yarn jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.6.jar  
pi 4 1000
```

Champ	Résultat
Application	application_1791039798429_0001
Nom	QuasiMonteCarlo
État final	SUCCEEDED
Estimation	$\pi \approx$ 3.14
Paramètres	4 maps, 1000 samples

```
root@hadoop-master:~# yarn application -list -appStates ALL
docker : 2026-10-04 01:03:01,304 INFO client.DefaultNoHARMFalloverProxyProvider: Connecting to ResourceManager at
hadoop-master/172.23.0.2:8032
Au caractère C:\Users\asiny\AppData\Local\Temp\ps-script-67860ea5-b239-4dd4-997c-850c8a40b275.ps1:107 : 9
+ $yarn = docker exec hadoop-master bash -c 'yarn application -list -ap ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (2026-10-04 01:0...172.23.0.2:String) [], RemoteException
+ FullyQualifiedErrorId : NativeCommandError

Total number of applications (application-types: [], states: [NEW, NEW_SAVING, SUBMITTED, ACCEPTED, RUNNING, FINISHED, FAILED, KILLED] and tags:
[]):1
Application-Id      Application-Name      Application-Type      User      Queue      State
Final-State
application_1791039798429_0001      QuasiMonteCarlo      MAPREDUCE      root      default      FINISHED
SUCCEEDED      100% http://hadoop-worker1:19888/jobhistory/job/job_1791039798429_0001
root@hadoop-master:~# yarn application -status application_1791039798429_0001 2>&1 | head -n 30
2026-10-04 01:03:05,625 INFO client.DefaultNoHARMFalloverProxyProvider: Connecting to ResourceManager at hadoop-master/172.23.0.2:8032
2026-10-04 01:03:06,463 INFO conf.Configuration: resource-types.xml not found
2026-10-04 01:03:06,464 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
Application Report :
  Application-Id : application_1791039798429_0001
  Application-Name : QuasiMonteCarlo
  Application-Type : MAPREDUCE
  User : root
  Queue : default
  Application Priority : 0
  Start-Time : 1791070566554
  Finish-Time : 1791070642461
  Progress : 100%
  State : FINISHED
  Final-State : SUCCEEDED
  Tracking-URL : http://hadoop-worker1:19888/jobhistory/job/job_1791039798429_0001
  RPC Port : 41245
  AM Host : hadoop-worker1
  Aggregate Resource Allocation : 218745 MB-seconds, 245 vcore-seconds
  Aggregate Resource Preempted : 0 MB-seconds, 0 vcore-seconds
  Log Aggregation Status : DISABLED
```

Capture terminal — YARN


FINIS

Cluster

About
Nodes
Node Labels
Applications
NEW
NEW SAVING
SUBMITTED
ACCEPTED
RUNNING
FINISHED
FAILED
KILLED
Scheduler

Tools

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Used R
1	0	0	1	0	<memory:0 B, vCores

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes
5	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation
Capacity Scheduler	[memory-mb (unit=Mi), vcores]	<memory:256, vCores:1>

Show 20 entries

ID	User	Name	Application Type	Application Tags	Queue	Application Priority	StartTime	LaunchTime	FinishTime
application_1791039798429_0001	root	QuasiMonteCarlo	MAPREDUCE		default	0	Sat Oct 3 19:36:06 -0400 2026	Sat Oct 3 19:36:11 -0400 2026	Sat Oct 3 19:37:22 -0400 2026

Showing 1 to 1 of 1 entries

UI YARN — applications


Application application_1791039798429_0001
Logged in as: dr:who

Cluster

About
Nodes
Node Labels
Applications
NEW
NEW SAVING
SUBMITTED
ACCEPTED
RUNNING
FINISHED
FAILED
KILLED
Scheduler

Tools

Application Overview

User: root

Name: QuasiMonteCarlo

Application Type: MAPREDUCE

Application Tags:

Application Priority: 0 (Higher Integer value indicates higher priority)

YarnApplicationState: FINISHED

Queue: default

FinalStatus Reported by AM: SUCCEEDED

Started: Sat Oct 03 23:36:06 +0000 2026

Launched: Sat Oct 03 23:36:11 +0000 2026

Finished: Sat Oct 03 23:37:22 +0000 2026

Elapsed: 1mins, 15sec

Tracking URL: History

Log Aggregation Status: DISABLED

Application Timeout (Remaining Time): Unlimited

Diagnostics:

Unmanaged Application: false

Application Node Label expression: <Not set>

AM container Node Label expression: <DEFAULT_PARTITION>

Application Metrics

Total Resource Preempted: <memory:0, vCores:0>

Total Number of Non-AM Containers Preempted: 0

Total Number of AM Containers Preempted: 0

Resource Preempted from Current Attempt: <memory:0, vCores:0>

Number of Non-AM Containers Preempted from Current Attempt: 0

Aggregate Resource Allocation: 210745 MB-seconds, 245 vcore-seconds

Aggregate Preempted Resource Allocation: 0 MB-seconds, 0 vcore-seconds

Show 20 entries

Search:

Attempt ID	Started	Node	Logs	Nodes blacklisted by the app	Nodes blacklisted by the system
appattempt_1791039798429_0001_000001	Sat Oct 3 19:36:06 -0400 2026	http://hadoop-worker1.8042	Logs	0	0

Showing 1 to 1 of 1 entries

First Previous 1 Next Last

Détail application SUCCEEDED

Partie 3 — Administration, monitoring et analyse (10 pts)

1. Interfaces web (4 pts)

NameNode — <http://localhost:9870>

Le cluster affiche **5 DataNodes** actifs. L'espace DFS utilisé reste faible (environnement de laboratoire). Aucun *missing block* observé pendant nos tests.

YARN — <http://localhost:8088>

L'application pi apparaît en état **FINISHED / SUCCEEDED**. Les métriques montrent l'allocation mémoire et le temps d'exécution (environ 198589 MB-seconds et 216 vcore-seconds sur un des runs).

2. Facteur de réplication (3 pts)

```
hdfs dfs -setrep -w 2 /data/ventes/2026/transactions.csv
```

Après la commande, la réplication du fichier passe à **2**. Le NameNode met à jour la cible. Comme on diminuait de 3 à 2, une copie de bloc superflu est retirée. L'option `-w` attend la fin de l'opération.

3. Retour d'expérience / troubleshooting (3 pts)

#	Problème	Cause	Solution
1	<code>pull access denied</code>	Compose téléchargeait une image pas encore sur Hub	<code>pull_policy: never</code> + build local
2	<code>repository name must be lowercase</code>	Majuscules dans le tag (<code>G4</code>)	Tag final <code>leshadoopridders/hadoop-tp-g4:1.0</code>
3	Build bloqué longtemps	Téléchargement trop lent depuis <code>archive.apache.org</code>	Base <code>apache/hadoop:3.3.6</code> + notre config / <code>entrypoint.sh</code>

Annexes

Structure du code source

```
Les_Hadoop_Riders_G4/
├── Dockerfile
├── docker-compose.yml
├── entrypoint.sh
├── config/
│   ├── core-site.xml
│   ├── hdfs-site.xml
│   ├── yarn-site.xml
│   ├── mapred-site.xml
│   └── workers
├── data/
│   ├── transactions.csv
│   └── transactions_part2.csv
├── scripts/
│   ├── demo-hdfs.sh
│   └── demo-yarn-pi.sh
├── captures/
└── README.md
```

Liens du projet

Ressource	URL
Docker Hub	https://hub.docker.com/r/leshadoopridders/hadoop-tp-g4
GitHub	https://github.com/asinyopetro/Les_Hadoop_Riders_G4
UI NameNode (local)	http://localhost:9870
UI YARN (local)	http://localhost:8088